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1 Introduction

2 Liquid-liquid phase separation

Discuss under what conditions a field theoretical description is valid

2.1 Thermodynamics of a ternary mixture

We begin by considering the a complex mixture of three components at constant temperature T, Volume V and particle numbers N_A, N_B, N_C. This system is best described by the Helmholtz free energy. $F = U - TS$. For simplicity, we only consider an incompressible system with constant volume and molecular volume ν. In this case a vacancy caused by removing one particle must be filled by another particle. Thus we can introduce the relationship

$$\phi_1 + \phi_2 + \phi_3 = 1 \tag{1}$$

where $\phi_i = \frac{N_i \nu}{V}$ is the volume fraction of component i. This relationship allows us to reduce the number of independent variables from three to two. Thus the free energy density can be written as $f(\phi_1, \phi_2)$.

2.1.1 Local stability analysis

In a homogenous system the total free energy is equal to $F = Vf(\phi_1, \phi_2)$. The local stability of this system is determined by the curvature of the free energy landscape. In particular the system becomes unstable when the Hessian matrix of the free energy with respect to the volume fractions has a vanishing determinant. This is when the curvature in one changes sign and the system can lower its free energy by phase separating. To better understand why phase separating lowers the free energy it helps to take a closer look at the properties of concave functions. In general a concave function f satisfies the property

$$f(\lambda x_1 + (1-\lambda)x_2) \geq \lambda f(x_1) + (1-\lambda) f(x_2) \quad \text{for } \lambda \in [0,1]$$

For a system inside of the spinodal region the free energy density is concave in some direction in the (ϕ_1, ϕ_2) plane, which implies

$$f(\bar{\boldsymbol{\phi}}) \geq \lambda f(\boldsymbol{\phi}^{(1)}) + (1-\lambda) f(\boldsymbol{\phi}^{(2)})$$

where we use the vector notation $\boldsymbol{\phi} = (\phi_1, \phi_2)^T$ and we have defined the average volume fraction $\bar{\boldsymbol{\phi}} = \lambda\boldsymbol{\phi}^{(1)} + (1-\lambda)\boldsymbol{\phi}^{(2)}$. This means that the system can lower its free energy by separating into two phases with volume fractions $\boldsymbol{\phi}^{(1)}$ and $\boldsymbol{\phi}^{(2)}$ and relative volumes λ and $1-\lambda$. Note that this only works for systems where the interface region between the phases is much smaller than the bulk phases, otherwise the interfacial energy needs to be taken into account. Since the spinodal is given by one condition it defines one or multiple lines in the (ϕ_1, ϕ_2) plane and separates locally stable from unstable regions.

2.1.2 Global stability analysis

Local stability does not imply global stability. Although a system does not spontaneously phase separate there might still exist a phase separated state with lower free energy. To determine whether such a state exists we can examine the free energy of such a phase separated state. Again neglecting the interfacial energy, the free energy of a two phase system is given by

$$V_1 f(\boldsymbol{\phi}^{(1)}) + V_2 f(\boldsymbol{\phi}^{(2)}) \tag{2}$$

where V_1 and V_2 are the volumes of phase 1 and 2 respectively and $\boldsymbol{\phi}^{(1)}$ and $\boldsymbol{\phi}^{(2)}$ are the corresponding volume fractions. The total volume and total number of particles need to be conserved, which yields additional equations $V_1\phi_i^{(1)} + V_2\phi_i^{(2)} = \bar{\phi}_i$ for $i = 1, 2$ and $V_1 + V_2 = V$. Therefore the six variables reduce to three independent variables. Minimizing the free energy in Eq. 2 with respect to these three variables yields the coexistence conditions

$$\begin{aligned}
\frac{df(\boldsymbol{\phi}^{(1)})}{d\phi_1} &= \frac{df(\boldsymbol{\phi}^{(2)})}{d\phi_1} \\
\frac{df(\boldsymbol{\phi}^{(1)})}{d\phi_2} &= \frac{df(\boldsymbol{\phi}^{(2)})}{d\phi_2} \\
\phi_i^{(1)}\frac{df(\boldsymbol{\phi}^{(1)})}{d\phi_i} - f(\boldsymbol{\phi}^{(1)}) &= \phi_i^{(2)}\frac{df(\boldsymbol{\phi}^{(2)})}{d\phi_i} - f(\boldsymbol{\phi}^{(2)})
\end{aligned}$$

Identifying the excess chemical potential $\mu_i = \frac{df}{d\phi_i}$ and osmotic pressure $\Pi = \phi_i\frac{df}{d\phi_i} - f$ we see that at coexistence the chemical potentials and osmotic

pressures of both phases need to be equal.

$$\mu_1(\boldsymbol{\phi}^{(1)}) = \mu_1(\boldsymbol{\phi}^{(2)}) \tag{3a}$$

$$\mu_2(\boldsymbol{\phi}^{(1)}) = \mu_2(\boldsymbol{\phi}^{(2)}) \tag{3b}$$

$$\Pi(\boldsymbol{\phi}^{(1)}) = \Pi(\boldsymbol{\phi}^{(2)}) \tag{3c}$$

The set of all points that satisfy these conditions define the binodal line in the (ϕ_1, ϕ_2) plane. Geometrically this condition can be interpreted as a common tangent construction in the free energy landscape. Add figure here In general the spinodal will always be enclosed by the binodal, since local instability implies global instability. Going the other way around is not necessarily true, since a system can be locally stable but globally unstable. This region between the binodal and spinodal lines is called metastable region. For biological systems this means that in order to reach the global minimum the system needs to overcome an energy barrier. This can happen in the presence of thermal fluctuations, which can push the system over the barrier. This process is called nucleation but we will not discuss it further in this text. When analysing properties of a system one can use the binodal and spinodal lines to approximate an upper and lower bound. As an example nucleation rates are bounded from below at the spinodal and from above at the binodal.

2.1.3 Critical point

The critical point is the endpoint of the binodal curve at which the two coexisting phases become identical. At this point the difference between the two equilibrium compositions vanishes continuously and the tie line shrinks to a single point in composition space. Consequently, the binodal and spinodal curves intersect at the critical point.

Since the critical point lies on the spinodal, the free energy must satisfy the same stability condition. In particular, the determinant of the Hessian vanishes,

$$\det\left(\frac{\partial^2 f}{\partial\phi_i\partial\phi_j}\right) = 0. \tag{4}$$

Geometrically, the critical point marks the transition between first- and second-order phase transition behavior along the coexistence curve. Away from the critical point, crossing the binodal leads to a discontinuous change in the equilibrium composition. The system separates into two phases with

finite differences in their volume fractions. This discontinuous change is characteristic of a first-order phase transition.

At the critical point, however, the difference between the coexisting phase compositions arises continuously. In this sense, the critical point represents a second-order phase transition in composition space.

Mathematically, the emergence of two minima from a single minimum in the free energy occurs precisely when the third derivative of the free energy vanishes along the unstable direction. Denoting by v_i the eigenvector corresponding to the vanishing eigenvalue of the Hessian, this condition can be written as

$$\sum_{i,j,k} v_i v_j v_k \frac{\partial^3 f}{\partial \phi_i \partial \phi_j \partial \phi_k} = 0. \tag{5}$$

Close to the critical point, the system exhibits universal scaling behavior that is independent of the microscopic details of the model. In particular, the difference between the coexisting phase compositions and other thermodynamic quantities follow power-law relations controlled by universal critical exponents. These scaling relations will be derived in the next chapter.

Add schematic figure of FH energy for binary system for $chi < 2$, $chi > 2$ and at the critical point $chi = 2$

2.2 Mean-field approximation of lattice interactions

In the previous sections we have discussed properties of the free energy and how these properties can drive phase separation. In this section we will focus on the exact form of the free energy. I could add the explicit FH free energy density here

2.2.1 Ising model

We start by considering a one dimensional lattice model with three possible states at each lattice site $1, 2, 3$ where the total number of each state is conserved. Because we do not allow for vacant lattice cites we can reduce the number of states from three to two due to the condition:

$$\sigma_1 + \sigma_2 + \sigma_3 = 1$$

Interactions between lattice sites are restricted to nearest neighbors and have interaction energies ϵ_{ij} where $i, j \in \{1, 2\}$. Note that since we are investigating the reduced system the interaction energies ϵ_{ij} have absorbed the

interactions with the third component.
The full Hamiltonian has the form:

$$H = \sum_{n}^{N} \epsilon_{11}\sigma_1^n\sigma_1^{n+1} + \epsilon_{22}\sigma_2^n\sigma_2^{n+1} + \epsilon_{12}(\sigma_1^n\sigma_2^{n+1} + \sigma_2^n\sigma_1^{n+1}) \tag{6}$$

Reducing the number of states introduces linear terms, but becuase the total number of each state is conserved these terms simply add a constant to the Energy and can thus be neglected.

2.2.2 Mean-field and continuum limit

Next we can rewrite the same-species and cross-species interactions in terms of single site and gradient expressions:

$$2\sigma_j^n\sigma_j^{n+1} = (\sigma_j^n)^2 + (\sigma_j^{n+1})^2 - (\sigma_j^{n+1} - \sigma_j^n)^2$$
$$\sigma_1^n\sigma_2^{n+1} + \sigma_2^n\sigma_1^{n+1} = \sigma_1^n\sigma_2^n + \sigma_1^{n+1}\sigma_2^{n+1} - (\sigma_1^{n+1} - \sigma_1^n)(\sigma_2^{n+1} - \sigma_2^n)$$

We will also introduce the mean field approximation such that we can replace the lattice sites and their gradients by their average values $\langle\sigma_j^n\rangle$ and $\langle\sigma_j^{n+1} - \sigma_j^n\rangle$ respectively. Lastly, we will take the continuum limit where the lattice spacing goes to zero, such that $\langle\sigma_j^n\rangle \to \phi_j(x)$ and $\langle\sigma_j^{n+1} - \sigma_j^n\rangle \to \nu^{1/3}\nabla\phi_j(x)$ where ν is the lattice volume . The internal Energy from Eq. 6 now has the form:

$$E = \nu^{-1/3}\int dx \frac{1}{2}\phi_i(x)\epsilon_{ij}\phi_j(x) - \frac{\nu^{2/3}}{2}\nabla\phi_i(x)\epsilon_{ij}\nabla\phi_j(x)$$

where we have introduced Einstein summation notation. The entropy can be derived from combinatoric arguments and is given by: Does this need a derivation, because I don't know how

$$\begin{aligned} TS = -\frac{k_BT}{\nu^{1/3}}\int dx[&\phi_1(x)\ln(\phi_1(x)) + \phi_2(x)\ln(\phi_2(x)) \\ &+ (1-\phi_1(x)-\phi_2(x))\ln(1-\phi_1(x)-\phi_2(x))] \end{aligned}$$

Therefore the free energy functional $F = E - TS$ has the form

$$\begin{aligned} F[\phi_1,\phi_2] = \frac{k_BT}{\nu^{1/3}}\int dx \frac{1}{2}\phi_i\chi_{ij}\phi_j + \frac{1}{2}\nabla\phi_i\kappa_{ij}\nabla\phi_j \\ + \phi_1\ln(\phi_1) + \phi_2\ln(\phi_2) + (1-\phi_1-\phi_2)\ln(1-\phi_1-\phi_2) \end{aligned} \tag{7}$$

Where $\chi_{ij} = \frac{1}{k_B T}\epsilon_{ij}$ is the Flory-Huggins parameter matrix and $\kappa_{ij} = -\frac{\nu^{2/3}}{k_B T}\epsilon_{ij}$ is the gradient coefficient matrix. The free energy density f is thus given by:

$$\begin{aligned}\frac{f(\phi_1, \phi_2)}{k_B T \nu^{-1/3}} = \frac{1}{2}\phi_i \chi_{ij} \phi_j + \frac{1}{2}\nabla\phi_i \kappa_{ij} \nabla\phi_j + \\ \phi_1 \ln(\phi_1) + \phi_2 \ln(\phi_2) + (1 - \phi_1 - \phi_2)\ln(1 - \phi_1 - \phi_2)\end{aligned} \tag{8}$$

Note that this is a line density and can be thought of as having absorbed the interfacial area. As we can see the Flory-Huggins parameters and gradient coefficients are directly related to the interaction energies of the lattice model and thus not independent parameters. At this point it is worth mentioning that there exists various approaches to derive the Flory-Huggins free energy that give different relationships between the Flory-Huggins parameters and the gradient coefficients and even depend on volume fractions. Biological systems are nuanced and likely more complex than any of these theoretical models can capture. The main purpose of this derivation is to illustrate that such a relationship can exist and to reduce the number of independent parameters.
In literature one will often find an additional parameter Ω_i which is the molecular size of the components in units of the lattice size. Difference in molecular size will change the combinatorics of the system and thus the entropy contribution to the free energy. We will not go into detail here since we assume all components to have the same molecular size.

2.2.3 Interface stability

Unlike the previous sections the free energy Eq. 7 contains gradient terms. These terms penalize sharp interfaces and lead to stable interface profiles between coexisting phases. One requirement for the κ matrix is that it needs to be positive definite. Otherwise the free energy could be lowered indefinitely by increasing gradients in certain directions. This condition translates to $\kappa_{11} > 0, \kappa_{22} > 0$ and $|\kappa| > 0$. Because of the relationship between κ and χ this also implies that $\chi_{11} < 0, \chi_{22} < 0$ and $|\chi| < 0|$. Further is has implications on the allowed values of the interaction parameters in the full system $\chi_{12}, \chi_{13}, \chi_{23}$. The null-void interaction matrix is given by:

$$\chi = \begin{pmatrix} -2\chi_{10} & \chi_{12} - \chi_{10} - \chi_{20} \\ \chi_{12} - \chi_{10} - \chi_{20} & -2\chi_{20} \end{pmatrix}$$

It is easy to see that $\chi_{10} > 0$ and $\chi_{20} > 0$ are required for $\chi_{11} < 0$ and $\chi_{22} < 0$. The last condition $|\chi| < 0$ translates to

$$\chi_{12} \in [(\sqrt{\chi_{10}} - \sqrt{\chi_{20}})^2, (\sqrt{\chi_{10}} + \sqrt{\chi_{20}})^2]$$

Thus all interaction parameter need to be positive, which may come as a surprise since negative interaction parameters are often used to model attractive interactions. One way to interpret this result is to look at the relative difference between interaction parameters. For example if χ_{10} and χ_{20} are large compared to χ_{12} component 1 and 2 will effectively attract each other since they both repel component 0 strongly. Additionally we will see that in our systems phase separation only occurs once the interaction parameter are above 2 such that the condition $\chi_{ij} > 0$ still allows for homogenous mixtures.

2.3 Dynamics of phase separation

In this section we will derive the laws that govern the dynamics of phase separation in ternary mixtures. We will start with the most general case and then make simplifications that are valid for the systems we are interested in. The results are all thermodynamically consistent and ensure positive entropy production.

2.3.1 Conservation Laws

Due to mass conservation each component satisfies a continuity equation of the form

$$\frac{\partial \phi_i}{\partial t} + \nabla \cdot \boldsymbol{J}_i = 0$$

Where $\boldsymbol{J}_i$ is the spacial flux of component i. The flux can be separated into an advective part due to the center of mass velocity field $\boldsymbol{v} = \sum_{i=0}^{N_c} \boldsymbol{J}_i$ and a relative flux $\boldsymbol{j}_i = \phi_i(\boldsymbol{v}_i - \boldsymbol{v})$ driven by chemical potential gradients, where $\boldsymbol{v}_i = \boldsymbol{J}_i/\phi_i$. The continuity equation can thus be rewritten as

$$\frac{\partial \phi_i}{\partial t} + \nabla \cdot (\phi_i \boldsymbol{v}) + \nabla \cdot \boldsymbol{j}_i = 0 \tag{9}$$

In the following we will restrict ourselves to systems where diffusion dominates over advection, i.e. the second term in Eq. 9 can be neglected. This is valid for systems with low Peclet number $Pe = \frac{Lv}{D} \ll 1$ where L is a characteristic length scale, v a characteristic velocity scale and D a characteristic

diffusion coefficient.
To find the relationship between the relative fluxes and chemical potential gradients we can look at the entropy production of the system. For a closed system at constant temperature and volume the change in entropy is related to the free energy change by $\frac{dS}{dt} = -\frac{1}{T}\frac{dF}{dt}$. Using the chain rule the time derivative of the free energy functional can be written as

$$\frac{dF}{dt} = \int d\boldsymbol{r} \frac{\delta f}{\delta \phi_i} \frac{\partial \phi_i}{\partial t} = \int d\boldsymbol{r} \bar{\mu}_i \frac{\partial \phi_i}{\partial t}$$

where we have identified the exchange chemical potential as the functional derivative of the free energy with respect to the volume fractions $\bar{\mu}_i = \frac{\delta F}{\delta \phi_i}$. Inserting the continuity equation and performing partial integration yields

$$\frac{dF}{dt} = \int d\boldsymbol{r} \nabla \bar{\mu}_i \cdot \boldsymbol{j}_i$$

where we have used that the fluxes vanish at the system boundaries. The entropy production is thus given by

$$\frac{dS}{dt} = -\frac{1}{T} \int d\boldsymbol{r} \nabla \bar{\mu}_i \cdot \boldsymbol{j}_i$$

which needs to be positive according to the second law of thermodynamics. The simplest linear relationship that ensures positive entropy production is given by

$$\boldsymbol{j}_i = -\sum_{i=1}^{N_c} \Lambda_{ij} \nabla \bar{\mu}_j \tag{10}$$

where Λ_{ij} is a positive definite mobility matrix of the reduced system. To due Onsager reciprocal relations the mobility matrix needs to be symmetric $\Lambda_{ij} = \Lambda_{ji}$. Inserting Eq. 10 into the continuity equation Eq. 9 yields the general form of the Cahn-Hilliard equations for a ternary mixture without hydrodynamics or thermal fluctuations:

$$\frac{\partial \phi_i}{\partial t} = \nabla \cdot \left(\sum_{j=1}^{N_c} \Lambda_{ij}(\boldsymbol{\phi}) \nabla \bar{\mu}_j \right) \tag{11}$$

where in general the mobility matrix can depend on the local volume fractions.

2.4 Surfacetension

The surfacetension is a thermodynamic quantity which is related to the energy per area of the interface between two coexisting phases. It plays an important role when considering hydrodynamic effects or nucleation phenomena. In order to compute the surface tension one must first set the interface profile between the two phases. For a binary system this is straightforward if one defines the inflection point of the profile as the interface location. Then the surface tension is simply $\gamma = F/A - x_I f(\phi^{(1)}) - (L - x_I) f(\phi^{(2)})$ where x_I is the inflection point and A the cross-sectional area of the system. Unfortunately a ternary system does not generally have this symmetry in its interface profiles and thus no clear way to define the interface location. This forces one to think about the surface tension as a system property, rather than a geometric property. Thus the surface tension should be the same quantity regardless of how we define the interface location. In the following we will present two different approaches to derive the surface tension of a ternary system, once with a thermodynamic argument and once with a mechanical argument.

2.4.1 Thermodynamic derivation

We start by dividing the system into two bulk and one interface region. The free energy, entropy and the particle number of each region are given by.

$$F = F^{(1)} + F^s + F^{(2)}$$
$$S = S^{(1)} + S^s + S^{(2)}$$
$$N_i = N_i^{(1)} + N_i^s + N_i^{(2)}$$

where s denotes the quantities assosiated with the interface and are known as excess quantitites. The work done by a systems is given by

$$dU = dQ + dW = TdS + \sum_i \mu_i dN_i - pdV + \gamma dA$$

where the first two terms are the reversible heat received by the system and the chemical heat generated by the system respectively. The last two terms are the mechanical work dW done on the system. Next we can write the total derivative of the free energy of the systems $F = U - TS$

$$dF = -SdT - pdV + \gamma dA + \sum_i \mu_i dN_i$$

The surface area A is an interfacial quantity, such that we can rewrite the last equation in terms of excess quantities:

$$dF^s = -S^s dT + \gamma dA + \sum_i \mu_i dN_i^s$$

notice that the volume is not an excess quantity. if we now consider a process where we increase the area for constant temperature, volume and chemical potentials we find

$$\gamma = \frac{1}{A}(F^s - \sum_i \mu_i \Gamma_i) \tag{12}$$

where $\Gamma_i = N_i - N_i^{(1)} - N_i^{(2)}$ is the excess particle number. Therefore the main difference between the ternary and the binary case is that in the ternary case we need to consider the excess particle number in addition to the excess free energy. As a matter of fact the expression for the binary case is recovered when we select an interface location such that the excess particle number vanishes for all species, which in general is not possible to find for a ternary system.

2.4.2 Mechanical derivation

2.5 Numerical methods

2.5.1 Euler-Lagrange formalism

In the previous sections we have derived the kinetics of phase separation from the general continuity equations and thermodynamic arguments. The resulting Cahn-Hilliard equations yield the equilibrium profiles by construction, but solving them numerically can be quite challenging due to the presence of fourth order derivatives. Instead one can also derive the equilibrium profiles from variational calculus by minimizing the free energy functional with respect to the volume fractions $\delta F/\delta\phi_i$ which yields the Euler-Lagrange equations:

$$\frac{\delta F[\phi_1, \phi_2]}{\delta \phi_i(x)} = \frac{\partial f(x)}{\partial \phi_i(x)} - \nabla \cdot \frac{\partial f(x)}{\partial \nabla \phi_i(x)} = 0$$

Inserting the Flory-Huggins free energy density from Eq. 8 yields a set of coupled second order PDEs for the volume fractions.

$$\begin{aligned} &\ln(\boldsymbol{\phi}) - \ln\big(1 - \mathbf{1}^\top \boldsymbol{\phi}\big)\,\mathbf{1} + \boldsymbol{\chi}\boldsymbol{\phi} - \boldsymbol{\kappa}\Delta\boldsymbol{\phi} = 0 \\ \Rightarrow \quad &\boldsymbol{\phi} = \boldsymbol{\chi}^{-1}\left(\boldsymbol{\kappa}\Delta\boldsymbol{\phi} - \ln(\boldsymbol{\phi}) + \ln\big(1 - \mathbf{1}^\top\boldsymbol{\phi}\big)\,\mathbf{1}\right) =: T(\boldsymbol{\phi}) \end{aligned}$$

Here and in the following, functions of vectors such as $\ln(\boldsymbol{\phi})$ are understood componentwise.

These equations can be solved using a fixed point iteration scheme. Since T is generally not contractive, the naive iteration $\boldsymbol{\phi}^{i+1} = T(\boldsymbol{\phi}^i)$ may oscillate or diverge. We therefore use an explicit Euler step in an artificial (pseudo-)time variable with $0 < h \leq 1$:

$$\boldsymbol{\phi}^{i+1}(x) = \boldsymbol{\phi}^i(x) + h\left[T(\boldsymbol{\phi}^i(x)) - \boldsymbol{\phi}^i(x)\right] = \boldsymbol{\phi}^i(x) + hG(\boldsymbol{\phi}^i(x))$$

where we have defined $G(\boldsymbol{\phi}) = T(\boldsymbol{\phi}) - \boldsymbol{\phi}$.
To enforce mass conservation of each component at each step we modify the iteration scheme to:

$$\boldsymbol{\phi}^{i+1}(x) = \boldsymbol{\phi}^i(x) + h\left[G(\boldsymbol{\phi}^i(x)) - \langle G(\boldsymbol{\phi}^i(x))\rangle_x\right] \tag{13}$$

In principle, mass conservation could be enforced by introducing Lagrange multipliers in the free energy functional. Here, we instead assume the dynamics of the associated multipliers to be instantaneous, which leads to the projection onto the zero-mean subspace in Eq. (13).
Since the fixed points of G coincide with those of T, the modified iteration converges to the same equilibrium solutions of the Euler–Lagrange equations. Although the resulting dynamics are artificial and do not represent the physical time evolution, the scheme nevertheless converges to the same equilibrium profiles as the Cahn-Hilliard equations,while requiring the solution of only second order PDEs instead of fourth order ones.

2.5.2 Pseudo-spectral method

To solve Eq. 8, we employ a pseudo-spectral method, which is particularly efficient for fields governed by linear differential operators. Unlike finite-difference or finite-element approaches, spectral methods evaluate spatial derivatives exactly in Fourier space, which reduces discretization error and allows high accuracy with relatively few grid points. As a result, pseudo-spectral schemes often exhibit faster convergence and improved numerical stability for problems dominated by diffusive terms.

The central idea of the pseudo-spectral method is to use the Fourier transform to convert spatial derivatives into simple multiplications in Fourier space. In particular, second-order derivatives can be expressed as

$$\partial_{xx}\boldsymbol{\phi} = \mathcal{F}^{-1}\left[-k^2\,\mathcal{F}[\boldsymbol{\phi}]\right],$$

where $\mathcal{F}$ denotes the Fourier transform and k the wavevector. It allows us to evaluate the diffusive term in the operator $G(\boldsymbol{\phi})$ efficiently in Fourier space, while nonlinear terms are computed in real space.

Since the pseudo-spectral method relies on Fourier transforms, periodic boundary conditions are required. These are implemented by mirroring the system at the right-hand boundary, ensuring that the dense phase is always surrounded by the dilute phase.

2.5.3 Semi-implicit stepping scheme

Solving PDEs using explicit schemes can be quite inefficient due to stability constraints on the time step size. For example the diffusion equation $\dot{\phi} = D\phi_{xx}$ requires a time step size of $\Delta t < \frac{\Delta x^2}{2D}$ for stability when using an explicit Euler scheme. The instability arises from the high wavevector modes. One trick of supressing these instabilities is to modify the discretized time derivate of $\dot{\phi}$ by:

$$\frac{1}{h}(\boldsymbol{\phi}^{i+1} - \boldsymbol{\phi}^i) \to \left(\frac{1}{h} - \lambda\nabla^2\right)(\boldsymbol{\phi}^{i+1} - \boldsymbol{\phi}^i)$$

where λ is a positive constant. Inserting this into the fixed point iteration scheme from Eq. 13 and rearranging yields:

$$\boldsymbol{\phi}^{i+1} = \boldsymbol{\phi}^i + \frac{h}{1 - h\lambda\nabla^2}\left[G(\boldsymbol{\phi}^i) - \langle G(\boldsymbol{\phi}^i)\rangle_x\right] \tag{14}$$

From this we can see that high wavevector modes are supressed since $\frac{1}{1+h\lambda k^2} \ll 1$ for large k. This allows us to take much larger time steps without running into stability issues. Note that for $h \to 0$ the semi-implicit scheme reduces to the explicit scheme.

3 Surface tension of phase interfaces in ternary mixtures

In this chapter, we investigate the surface tension of interfaces in ternary mixtures composed of three distinct components (labeled 0, 1, and 2). Among these, component 2 serves as an interfacial modulator, characterized by the weakest repulsive interactions with the other two species. This property allows component 2 to preferentially accumulate at phase interfaces, effectively

mediating the interactions between components 0 and 1. We systematically vary the mean concentration of component 2 from a nearly binary limit to the critical point, examining how this modulation affects interfacial properties. We solve the interfacial problem numerically using the computational framework established in the previous section and additionally develop an analytical approximation that captures the essential physics. We then compare the analytical results to the numerical solutions to assess the accuracy and limitations of our theoretical approach. Finally, we derive the critical scaling behavior of the surface tension near the critical point.

3.1 Determening the interfacial profiles and surface tension numerically

3.1.1 Model setup and numerical scheme

After having rescaled the free energy density in Eq. 8 by $k_B T \nu^{-1/3}$ the model only depends on the Flory-Huggins interaction matrix χ_{ij} and the gradient prefactor λ. Since the gradient prefactor only sets the interfactial length scale, we fix $\lambda = 1$. With the convention used in Eq. (8), the gradient coefficient matrix is then

$$\kappa_{ij} = -\chi_{ij}. \tag{15}$$

In the following, we restrict attention to interaction parameters $\chi_{ij} \in [0, 3]$. This range reliably produces liquid–liquid phase separation, while avoiding (as far as possible) regions of three-phase coexistence; see the phase-diagram discussion in Sec. **??**.

For a chosen χ-matrix, we initialize the system using the corresponding coexisting bulk compositions. To reduce computational cost, these bulk phases are computed first (using the `flory` package) and then used as boundary values for the interfacial calculation. We initialize the interface as a sharp step between the two coexisting phases.

The spatial discretization is chosen adaptively to resolve the interface. The grid spacing Δx is set such that the interfacial region spans approximately 40 grid points. The total domain size is 256 grid points such that the fields relax to their bulk values well before the domain edges.

3.1.2 Equilibration dynamics and interfacial structure

Starting from the initial step profile, we relax the system to equilibrium using the fixed-point iteration in Eq. (13). Since we are only interested in the equilibrium profiles, the step size h is chosen as large as possible while maintaining numerical stability. The semi-implicit update strongly suppresses high-wavenumber modes, which allows us to use $h = \mathcal{O}(10^{-1})$, typically one to two orders of magnitude larger than in a fully explicit scheme.

We declare convergence when the maximum change in the volume fractions satisfies

$$\max_x \|\boldsymbol{\phi}^{i+1}(x) - \boldsymbol{\phi}^i(x)\| < 10^{-10}.$$

Depending on interaction parameters and initialization, this criterion is reached after 10^3–10^4 iterations.

We continue equilibrating the system for increased mean concentrations of the second species until the critical point beyond which two-phase coexistence no longer is feasible.

At equilibrium, the interface exhibits a smooth, sigmoid-like transition in each component. One the species with the weakest repulsive interactions with the other two components (i.e. the strongest effective attraction) is enriched at the interface, thus effectively screening the strongest repulsive interactions. Interestingly the degree of absorption at the interface increases as the interactions with the other two components becomes more repulsive and can even lead to a third phase. For a symmetric case $\chi_{01} = \chi_{02}$, the zeroth species is symmetric about the interface. In this regime, Huang and Olvera de la Cruz have shown that the interfacial adsorption scales linearly with the ratio χ_{01}/χ_{12}, matching our simulation data.

3.1.3 Critical scaling of the surface tension

Once the equilibrium profiles are obtained, the surface tension is computed using the thermodynamic relation derived in the previous section. The resulting surface tensions are shown in Fig. **??**.

As the system approaches the critical point, the interface continuously vanishes, which explains why the surface tension vanishes as well.

Close to criticality, we observe a clear power-law dependence of the surface tension on the distance from the critical point, which we quantify by the chemical potential difference $\mu_2 - \mu_{2,c}$:

$$\gamma \sim (\mu_2 - \mu_{2,c})^\delta.$$

This scaling is observed for a wide range of interaction parameters. By fitting the numerical data, we extract the corresponding critical exponent δ for each parameter set. The results are summarized in Fig. **??**.

Within numerical uncertainty, the critical exponent is universal across all interaction parameters considered and takes a value $\delta \approx 3/2$. This is consistent with the classical mean-field prediction for binary mixtures derived by Cahn [**?**]. In contrast, the prefactor of the power law is not universal and varies substantially with the choice of interaction parameters, reflecting non-universal microscopic details of the system.

Remarkably, the observed power-law behavior persists over a broad range of $\mu_2 - \mu_{2,c}$ values.

3.2 An effective interfacial model to approximate the surface tension

While the critical scaling of the surface tension near the critical point provides important physical insight, obtaining a closed-form expression for γ from the full ternary Flory-Huggins free energy is generally intractable. Consequently, for each choice of interaction parameters and mean concentration of the second species, the surface tension must be computed by solving the full numerical interfacial problem.

To avoid this computational cost, we now introduce an analytical approximation that estimates the surface tension using only the bulk volume fractions of the coexisting phases and the interaction parameters. The approximation remains quantitatively accurate over a broad range of interacrtion parameters and has the correct scaling near the critical point, as we will demonstrate later.

3.2.1 Derivation of a piecewise-linear interface

As shown in the previous sections, interfacial profiles in ternary mixtures can be highly nontrivial. To make progress, we approximate the interface by a piecewise-linear profile that connects the two coexisting bulk phases over a finite interfacial width.

This approximation neglects two important features of the exact solution. First, it assumes a common interfacial width for all components, which excludes adsorption of any species at the interface. Second, it ignores the intrinsic nonlinearity of the equilibrium profiles. Despite these simplifications,

the piecewise-linear ansatz allows for a closed-form expression for the surface tension.

To begin, we note that for a one-dimensional, planar interface at equilibrium, the Euler–Lagrange equations imply the existence of a conserved quantity across the interface. Writing the free-energy density as a sum of a local and a gradient contribution and adopting Einstein summation convention, we define

$$f(x) = f_{\text{local}}(\{\phi_i(x)\}) + \frac{1}{2}\,\nabla\phi_i(x)\,\kappa_{ij}\,\nabla\phi_j(x). \tag{16}$$

Introducing the exchange chemical potentials μ_i and osmotic pressure Π, the combination

$$f_{\text{local}}(\{\phi_i(x)\}) - \mu_i\phi_i(x) + \Pi - \frac{1}{2}\,\nabla\phi_i(x)\,\kappa_{ij}\,\nabla\phi_j(x) =: -V(\{\phi_i\}) - T(\{\nabla\phi_i\}) \tag{17}$$

is invariant across the interface. Here, the terms V and T play roles analogous to potential and kinetic energy in classical mechanics.

Taking the spatial derivative of Eq. (17) yields

$$\begin{aligned}\frac{d}{dx}\Big(f_{\text{local}}(\{\phi_i\}) - \mu_i\phi_i + \Pi - \frac{1}{2}\,\nabla\phi_i\,\kappa_{ij}\,\nabla\phi_j\Big) &= \left(\frac{\partial f_{\text{local}}}{\partial\phi_i} - \mu_i - \kappa_{ij}\Delta\phi_j\right)\frac{d\phi_i}{dx}\\ &= 0,\end{aligned}$$

where the expression in parentheses vanishes by virtue of the Euler–Lagrange equations. Consequently, the quantity in Eq. (17) is constant throughout the interfacial region.

For our approximation we assume that each component varies linearly across an interface of width w, connecting the bulk values $\phi_i^{(1)}$ and $\phi_i^{(2)}$. The magnitude of the gradient is therefore approximated by

$$|\nabla\phi_i| \simeq \frac{|\Delta\phi_i|}{w},$$

where $\Delta\phi_i = \phi_i^{(2)} - \phi_i^{(1)}$ denotes the difference between the coexisting bulk phases. The profile is centered around the mean composition

$$\bar{\phi}_i = \tfrac{1}{2}\left(\phi_i^{(1)} + \phi_i^{(2)}\right),$$

at which the local free-energy density is evaluated. Since we assume a common interfacial width for all components, this approximation holds for all

i. Substituting the gradient approximation into the quadratic gradient term yields

$$\nabla\bar{\phi}_i\,\kappa_{ij}\,\nabla\bar{\phi}_j = w^{-2}\,\Delta\phi_i\,\kappa_{ij}\,\Delta\phi_j.$$

Solving for the interfacial width gives

$$w = \sqrt{\frac{\Delta\phi_i\,\kappa_{ij}\,\Delta\phi_j}{\nabla\bar{\phi}_i\,\kappa_{ij}\,\nabla\bar{\phi}_j}}.$$

The denominator can be expressed using the invariant quantity Eq. (17), evaluated at the center of the interface. This leads to the approximate expression

$$w = \sqrt{\frac{\Delta\phi_i\,\kappa_{ij}\,\Delta\phi_j}{2\left(f_{\text{local}}(\{\bar{\phi}_i\}) - \mu_i\bar{\phi}_i + \Pi\right)}} \tag{18}$$

The surface tension corresponding to the piecewise-linear interfacial profile is obtained by integrating the gradient contribution to the free energy,

$$\gamma = \int dx\,\nabla\phi_i\,\kappa_{ij}\,\nabla\phi_j.$$

Within the linear-profile approximation, the gradients are constant across the interface and vanish in the bulk. The integral therefore reduces to

$$\gamma = w\,\nabla\bar{\phi}_i\,\kappa_{ij}\,\nabla\bar{\phi}_j,$$

Substituting the expression for w from Eq. (18) yields the approximate surface tension

$$\gamma = \sqrt{2\left(\Delta\phi_i\,\kappa_{ij}\,\Delta\phi_j\right)\left(f_{\text{local}}(\{\bar{\phi}_i\}) - \mu_i\bar{\phi}_i + \Pi\right)} \tag{19}$$

As anticipated, this expression depends only on the bulk compositions $\{\phi_i^{(1)}, \phi_i^{(2)}\}$ of the coexisting phases and the interaction parameters through κ_{ij}. Importantly, the closed-form nature of Eq. (19) allows derivatives of the surface tension with respect to order parameters to be evaluated analytically.

3.2.2 Comparison between analytical approximation and numerical results

The quantitative comparison between the analytical approximation and the numerical results is shown in Fig. 1. Somewhat surprisingly, the approximate

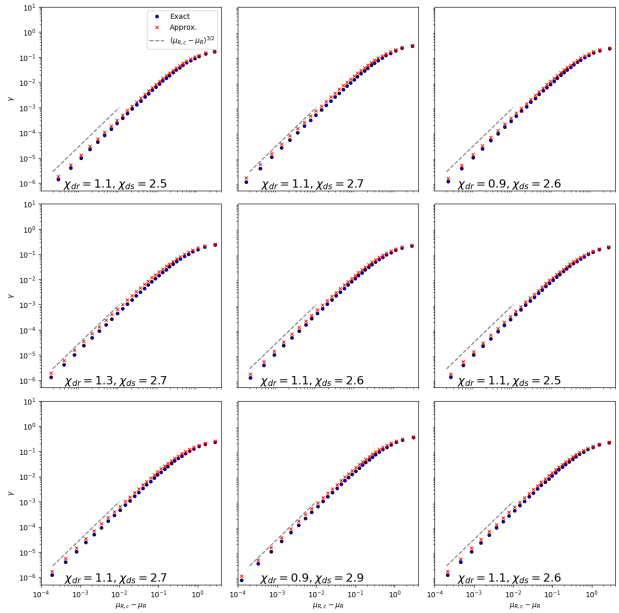

Figure 1: Exact (blue) and approximate (red) surface tension as a function of the chemical potential for various interaction parameters. For reference, the scaling behavior with exponent 1.5 is shown as a dashed line.

expression not only reproduces the correct critical scaling but also provides good quantitative agreement with the numerically computed surface tension over a wide range of parameters. Depending on the interaction matrix the surface tension will vary 10% to 70% from the exact value.

A systematic deviation is nevertheless apparent: the analytical approximation consistently overestimates the exact surface tension. This behavior can be understood naturally within the variational formulation of the interfacial problem. Using the invariant quantity from Eq. (17), the effective Lagrangian density of the system can be identified as

$$L = T - V = 2T = \nabla\phi_i \, \kappa_{ij} \, \nabla\phi_j,$$

where T and V denote the gradient and local contributions, respectively. The surface tension can therefore be interpreted as the action associated with the interfacial profile,

$$\gamma = \int dx \, L \equiv S.$$

Since the true equilibrium interface minimizes this action, any approximate interfacial profile necessarily yields a larger value of γ. The systematic overestimation observed in the analytical approximation is thus a direct consequence of the underlying variational principle.

3.2.3 Asymptotic critical scaling

Either use +/- or 1/2 for dense/dilute phase but not both. Also the derivation would benefit from a schematic figure showing the tie line, critical point and deviations in phi space.

In the previous section, we showed that the surface tension can be expressed approximately in terms of the bulk compositions of the coexisting phases. We now analyze the behavior of this expression in the vicinity of the critical point, with the goal of deriving its asymptotic scaling form.

Recall that the approximate surface tension is given by Eq. (19) as the product of two factors. We first consider the term

$$\tfrac{1}{2}\,\Delta\phi_i\,\kappa_{ij}\,\Delta\phi_j,$$

which depends solely on the difference between the bulk compositions of the two coexisting phases.

Close to the critical point, the volume fractions in the dense and dilute phases can be written as small deviations from the critical composition $\phi_{i,c}$,

$$\phi_i^{(1)} = \phi_{i,c} + \tfrac{1}{2}\Delta\phi_i, \quad \phi_i^{(2)} = \phi_{i,c} - \tfrac{1}{2}\Delta\phi_i,$$

where $\Delta\phi_i$ parameterizes the displacement of the coexisting phases along the tie line in composition space.

Expanding the chemical potentials in the two coexisting phases about the critical point yields

$$\boldsymbol{\mu}^{(1/2)} - \boldsymbol{\mu}_c = \boldsymbol{H}\,\Delta\boldsymbol{\phi}^{(1/2)} + \mathcal{O}(\Delta\phi^2), \tag{20}$$

where $\boldsymbol{H}$ denotes the Hessian of the local free-energy density evaluated at the critical composition, and $\Delta\boldsymbol{\phi}^{(1/2)} = \boldsymbol{\phi}^{(1/2)} - \boldsymbol{\phi}_c$. At coexistence, the chemical potentials in the two phases must be equal, $\boldsymbol{\mu}^{(1)} = \boldsymbol{\mu}^{(2)}$. Subtracting the two expansions and defining

$$\Delta\boldsymbol{\phi} = \Delta\boldsymbol{\phi}^{(1)} - \Delta\boldsymbol{\phi}^{(2)},$$

we obtain, to leading order,

$$\boldsymbol{H}\,\Delta\boldsymbol{\phi} = 0. \tag{21}$$

This shows that the difference between the bulk compositions of the coexisting phases lies in the null space of the Hessian at the critical point.

Furthermore, in the immediate vicinity of the critical point, the binodal is approximately linear. Consequently, the deviations $\Delta\boldsymbol{\phi}^{(1)}$ and $\Delta\boldsymbol{\phi}^{(2)}$ are parallel to $\Delta\boldsymbol{\phi}$ and thus also belong to the null space of $\boldsymbol{H}$. As a result, the linear term in the Taylor expansion of the chemical potentials vanishes along the coexistence direction, and higher-order terms control the leading behavior near criticality.

We now turn to the leading nonvanishing contribution in the Taylor expansion of the chemical potential about the critical point. Expanding to second order yields

$$\boldsymbol{\mu}^{(1/2)} - \boldsymbol{\mu}_c = \frac{1}{2}\left(\alpha^{(1/2)}\right)^2 \left[\frac{\partial^2 \mu_i}{\partial\phi_j \partial\phi_k}\,\Delta\phi_j \Delta\phi_k\right] \boldsymbol{e}_i + \mathcal{O}(\Delta\phi^3), \tag{22}$$

where all derivatives are evaluated at the critical composition, and we have introduced

$$\alpha^{(1)} = \frac{|\Delta\boldsymbol{\phi}^{(1)}|}{|\Delta\boldsymbol{\phi}|}, \quad \alpha^{(2)} = \frac{|\Delta\boldsymbol{\phi}^{(2)}|}{|\Delta\boldsymbol{\phi}|}.$$

Since the chemical potentials must be equal in the two coexisting phases, $\boldsymbol{\mu}^{(1)} = \boldsymbol{\mu}^{(2)}$, the prefactors must satisfy $\left(\alpha^{(1)}\right)^2 = \left(\alpha^{(2)}\right)^2$. Additionally the two phases must have displacements with opposite sign along the tie line, which implies

$$\Delta\boldsymbol{\phi}^{(1)} = -\,\Delta\boldsymbol{\phi}^{(2)} = \tfrac{1}{2}\,\Delta\boldsymbol{\phi}.$$

The terms inside of the brackets in Eq. (22) is generically nonzero. Consequently, the leading dependence of the chemical potential deviation on the distance from the critical point is

$$|\boldsymbol{\mu} - \boldsymbol{\mu}_c| \sim |\Delta\boldsymbol{\phi}|^2.$$

[1]

Finally, since $\Delta\boldsymbol{\phi}$ lies in the null space of the Hessian at the critical point, the leading nontrivial contribution to the dot product $(\boldsymbol{\mu} - \boldsymbol{\mu}_c)^T \Delta\boldsymbol{\phi}$ is equal

[1] The chemical potential remains constant along the tie line; therefore, it only depends on the magnitude $|\Delta\boldsymbol{\phi}|$ and not on the direction in composition space.

to the third derivative of the free energy. This quantity vanishes exactly at the critical point, implying that $\boldsymbol{\mu} - \boldsymbol{\mu}_c$ is orthogonal to the tie lines in (ϕ_1, ϕ_2) composition space.

We now turn to the second factor in the expression for the surface tension,

$$\nabla\bar{\phi}_i\,\kappa_{ij}\,\nabla\bar{\phi}_j = f(\bar{\boldsymbol{\phi}}) - \mu_i\bar{\phi}_i + \Pi = f(\bar{\boldsymbol{\phi}}) - f(\boldsymbol{\phi}^{(1)}) + \mu_i\big(\phi_i^{(1)} - \bar{\phi}_i\big), \tag{23}$$

where, in the first equality, we have used the conservation relation Eq. (17).

Close to the critical point, the dense-phase composition can be written as

$$\phi_i^{(1)} = \phi_{i,c} + \tfrac{1}{2}\,\Delta\phi_i,$$

while the midpoint composition satisfies

$$\bar{\phi}_i = \tfrac{1}{2}\big(\phi_i^{(1)} + \phi_i^{(2)}\big) = \phi_{i,c},$$

where the last equality follows from the symmetry $\Delta\boldsymbol{\phi}^{(1)} = -\Delta\boldsymbol{\phi}^{(2)}$.

We now expand the local free-energy density about the critical composition $\boldsymbol{\phi}_c$. Using Einstein summation convention, we obtain

$$\begin{aligned} f(\boldsymbol{\phi}^{(1)}) = f(\boldsymbol{\phi}_c) &+ \frac{1}{2}\,\mu_{i,c}\,\Delta\phi_i + \frac{1}{2^2\,2!}\,\Delta\phi_i\,\underbrace{H_{ij}\,\Delta\phi_j}_{=0} \\ &+ \frac{1}{2^3\,3!}\,\underbrace{\frac{\partial^3 f}{\partial\phi_i\partial\phi_j\partial\phi_k}\Delta\phi_i\Delta\phi_j\Delta\phi_k}_{=0} \\ &+ \frac{1}{2^4\,4!}\,\frac{\partial^4 f}{\partial\phi_i\partial\phi_j\partial\phi_k\partial\phi_l}\Delta\phi_i\Delta\phi_j\Delta\phi_k\Delta\phi_l + \mathcal{O}(\Delta\phi^5). \end{aligned} \tag{24}$$

The quadratic term vanishes because $\Delta\boldsymbol{\phi}$ lies in the null space of the Hessian at the critical point. The cubic term vanishes because the third derivative of the free energy along the coexistence direction is zero at criticality. Consequently, the leading nonzero contribution to the second factor in Eq. (23) arises from the quartic term, which is generically finite.

Finally, we revisit the higher-order expansion of the chemical potential. As shown earlier, the first-order contribution vanishes, and the second-order term is orthogonal to $\Delta\boldsymbol{\phi}$ and the zeroth term will cancel. Therefore we will include the third-order term as the first nonzero contribution in the second product. The chemical potential can then be written as

$$\tfrac{1}{2}\mu_i\Delta\phi_i = \tfrac{1}{2}\mu_{i,c}\Delta\phi_i + \frac{1}{2^4\cdot 3!}\,\frac{\delta^4 f}{\delta\phi_i\delta\phi_j\delta\phi_k\delta\phi_l}\,\Delta\phi_j\Delta\phi_k\Delta\phi_l + \mathcal{O}(\Delta\phi^5). \tag{25}$$

Collecting terms, we find

$$f(\boldsymbol{\phi}_c) - f(\boldsymbol{\phi}^{(+)}) + \tfrac{1}{2}\,\mu_i \Delta\phi_i = \frac{1}{2^7}\,\frac{\delta^4 f}{\delta\phi_i \delta\phi_j \delta\phi_k \delta\phi_l}\,\Delta\phi_i \Delta\phi_j \Delta\phi_k \Delta\phi_l + \mathcal{O}(\Delta\phi^5). \tag{26}$$

This shows that Eq. (23) scales as

$$|\Delta\boldsymbol{\phi}|^4 \;\propto\; |\boldsymbol{\mu} - \boldsymbol{\mu}_c|^2.$$

Combining this result with the scaling of the first factor, we conclude that near the critical point the surface tension obeys

$$\gamma \;\propto\; |\boldsymbol{\mu} - \boldsymbol{\mu}_c|^{3/2}.$$

In addition, this analysis provides the scaling of the interfacial width:

$$w = \frac{\sqrt{\Delta\phi_i\,\kappa_{ij}\,\Delta\phi_j}}{\sqrt{\nabla\bar{\phi}_i\,\kappa_{ij}\,\nabla\bar{\phi}_j}} \;\propto\; |\boldsymbol{\mu} - \boldsymbol{\mu}_c|^{-1/2}.$$

Which diverges at the critical point, as expected.

Taken together, these results provide a direct connection between the bulk thermodynamic quantities introduced in the previous chapter and the interfacial properties analyzed here. The critical scaling of the interfacial width $w \sim |\mu - \mu_c|^{-1/2}$ agrees with the diverging interfacial width observed in the numerical results, while the critical exponent of the surface tension $\gamma \sim |\mu - \mu_c|^{3/2}$ is consistent with previous work on critical interfacial behavior in mean-field models (cite work here). In this sense the effective interfacial model does not introduce new critical behavior, but rather implies that near the critical point the interfacial properties do not matter so much on the profile shape as they do on the bulk thermodynamics.

3.2.4 Sources of error and limitations

The piecewise-linear approximation differs from the true equilibrium interfacial profiles in two essential ways. First, it replaces the smooth, nonlinear variation of the volume fractions across the interface by a linear interpolation. Second, it assumes that all species share a common interfacial width, thereby neglecting preferential adsorption.

As shown in Fig. 1, the approximation performs best far from the critical point, where the mean concentration of one of the species nearly vanishes

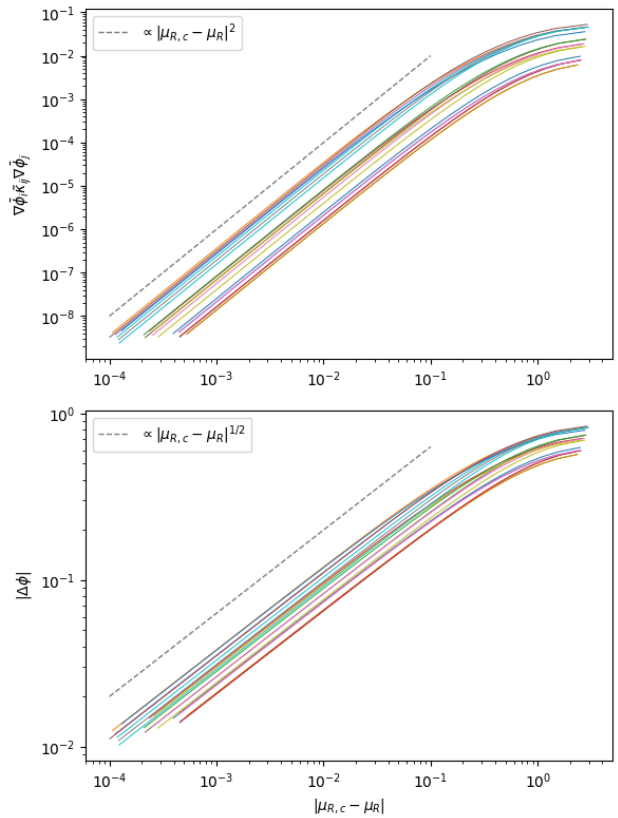

Figure 2: Numerical scaling behavior of the first (bottom) and second (top) product terms in Eq. 19, as a function of the chemical potential.

and the system effectively reduces to a binary mixture. In this regime, the two remaining components indeed exhibit nearly identical interfacial widths, and the dominant source of error arises from the simplified functional form of the profile. If one assumes that the error due to the wrong functional form varies only weakly across phase space, this suggests that the deviations from the equal-width assumption make up the primary source of error closer, in particular as one approache the critical point.

To quantify the importance of species dependent interfacial widths, we fit a hyperbolic tangent profile of the form

$$\phi(x) = \frac{\Delta\phi}{2} \tanh\left(\frac{x - x_0}{w}\right) + \bar{\phi}$$

to the numerical equilibrium profiles of both component 1 and component 0. The difference between the fitted interface positions x_0 for the two species then provides a direct measure of the deviation from the equal-width assumption and thus of the degree of interfacial adsorption.

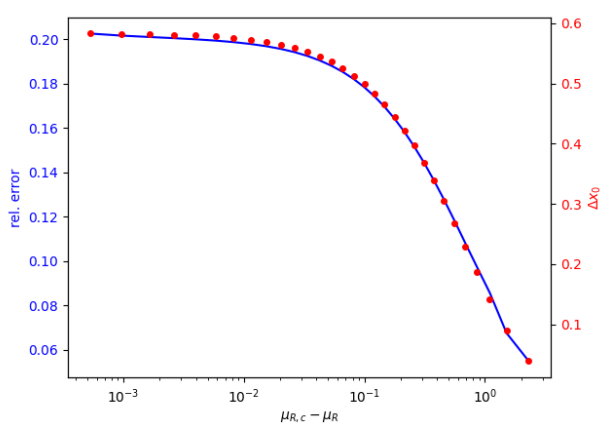

Figure 3: Displacement between the centers of the profiles of component 1 and component 0 (red) and the relative error $2(\gamma_{\text{approx}} - \gamma_{\text{exact}})/(\gamma_{\text{approx}} + \gamma_{\text{exact}})$ (blue) as a function of the chemical potential, for $\chi_{12} = 1.3$ and $\chi_{02} = 2.3$.

The results are shown in Fig. 3, together with the relative error of the surface-tension approximation. A clear correlation is observed between the relative error and the displacement between the centers of the profiles of the

first two components. profiles. Importantly, this correlation is not restricted to the specific interaction parameters shown in Fig. 3, but persists across a broad range of interaction strengths. This indicates that interfacial adsorption, rather than the simplified functional form of the profile, is the primary limitation of the piecewise-linear model.

3.2.5 Outlook and extensions

3.3 Analytical expressions for interfacial profiles in symmetric ternary mixtures

Although an analytical solution for the interfacial profile is generally not available, it is possible to derive an implicit expressions for the profile properties, for the symmetric case i.e. $\chi_{01} = \chi_{02}$ and $\phi_0^{(1)} = \phi_0^{(2)}$.

3.3.1 Implicit expression for the bulk phase compositions

The interaction matrix of the reduced two-component system is then given by:

$$\begin{pmatrix} \chi_d & \chi_o \\ \chi_o & \chi_d \end{pmatrix}$$

where χ_d, χ_o are the diagonal and off-diagonal elements, respectively. and $\kappa_{ij} = -\chi_{ij}$. The profiles are then given by the variational principle $\delta F/\delta\phi_i = 0$, which for the Flory-Huggins free energy in Eq. 7 leads to the following Euler–Lagrange equations:

$$\begin{aligned} -\chi_d\phi_1''(x) - \chi_o\phi_2''(x) &= \chi_d\phi_1 + \chi_o\phi_2 + \ln(\phi_1) - \ln(1-\phi_1-\phi_2) - \mu_1 \\ -\chi_o\phi_1''(x) - \chi_d\phi_2''(x) &= \chi_o\phi_1 + \chi_d\phi_2 + \ln(\phi_2) - \ln(1-\phi_1-\phi_2) - \mu_2 \end{aligned}$$

Where $\mu_1 = \mu_2 = \mu = \chi_d\phi_1^{(1)} + \chi_o\phi_2^{(1)} + \ln(\phi_1^{(1)}) - \ln(\phi_0^{(1)})$ is the chemical potential of the coexisting phases. The EL equations can be rewritten in terms of the eigenmodes of the interaction matrix $\phi_\pm = \frac{1}{2}(\phi_1 \pm \phi_2)$:

$$\begin{aligned} \frac{2}{\phi_s}\phi_+''(x) &= -\frac{2}{\phi_s}\phi_+ + \ln(\phi_+ + \phi_-) + \ln(\phi_+ - \phi_-) - 2\ln(1-2\phi_+) - 2\mu \\ \frac{2}{\bar{\phi}_c}\phi_-''(x) - &= -\frac{2}{\bar{\phi}_c}\phi_- + \ln(\phi_+ + \phi_-) - \ln(\phi_+ - \phi_-) \end{aligned}$$

Where we define $\chi_o - \chi_d =: 1/\bar{\phi}_c$ and $\chi_o + \chi_d =: -1/\phi_s$.
The chemical potential can be rewritten in terms of the mean volume fraction $\bar{\phi} = (\phi_1^{(1)} + \phi_2^{(1)})/2$ and the difference $\Delta\phi = (\phi_1^{(1)} - \phi_2^{(1)})/2$

$$\mu = -\bar{\phi}/\phi^* - \Delta\phi/\phi_c + \ln(\bar{\phi} + \Delta\phi) - \ln(1 - 2\bar{\phi})$$

. First we will focus on the EL equation for ϕ_-. For this we can look at the equation in the bulk where the laplacian is zero and $\phi_+ = \bar{\phi}$ is constant. Then we get the implicit equation for the bulk value of $\phi_-(x \to \infty) = \Delta\phi$:

$$\begin{aligned} &-\frac{2}{\bar{\phi}_c}\Delta\phi + \ln(\bar{\phi} + \Delta\phi) - \ln(\bar{\phi} - \Delta\phi) = 0 \\ \Rightarrow \quad &\Delta\phi = \bar{\phi}\tanh\left(\frac{\Delta\phi}{\phi_c}\right) \end{aligned} \tag{27}$$

3.3.2 Implicit expression for the interfacial absorption

Next we will look at the EL equation for ϕ_+ at the interface. Due to anti-symmetry of ϕ_- around the interface center we have $\phi_-(x = 0) = 0$ and therefore

$$\begin{aligned} &\frac{2}{\phi_s}\phi_+''(x) = -\frac{2}{\phi_s}\phi_+ + 2\ln(\phi_+) - 2\ln(1 - 2\phi_+) - 2\mu \\ \Rightarrow \quad &\frac{\phi_+''(x)}{\phi_s} = -\frac{\delta\phi_+}{\phi_s} + \frac{\Delta\phi}{\phi_c} + \ln\left(\frac{\bar{\phi} + \delta\phi_+}{\bar{\phi} + \Delta\phi}\right) + \ln\left(\frac{1 - 2\bar{\phi}}{1 - 2(\bar{\phi} + \delta\phi_+)}\right) \end{aligned}$$

Where we have defined $\delta\phi_+ = \phi_+ - \bar{\phi}$. Now we can rearrange this to the implicit equation for $\delta\phi_+$:

$$\delta\phi_+ = \frac{1 - 2\bar{\phi}}{2}\left(1 - \frac{1}{1 - 2\bar{\phi} + 2(\bar{\phi} + \Delta\phi)\Pi}\right) \tag{28}$$

Where we have defined

$$\Pi = \exp\left(\frac{\delta\phi_+}{\phi_s}(1 + a_2) - \frac{\Delta\phi}{\phi_c}\right) \tag{29}$$

where $a_2 = \delta\phi_+''(x)/\delta\phi_+$ measures the curvature of ϕ_+ at the interface center. Near the critical point the interface width w diverges. Consequently the order

parameter varies only over the large length scale w. Dimensional analysis then implies

$$\phi_+''(x) \sim \frac{\delta\phi_+}{w^2}$$

Since $w \to \infty$ as the critical point is approached, the ratio

$$a_2 = \frac{\phi_+''(x)}{\delta\phi_+} \sim \frac{1}{w^2}$$

vanishes in this limit. The curvature term therefore becomes parametrically small compared to the local terms in the Euler–Lagrange equation, justifying the approximation $a_2 \ll 1$. Under this assumption the equation for $\delta\phi_+$ can be solved self-consistently once $\Delta\phi$ is known. The results can be seen in figure 4.

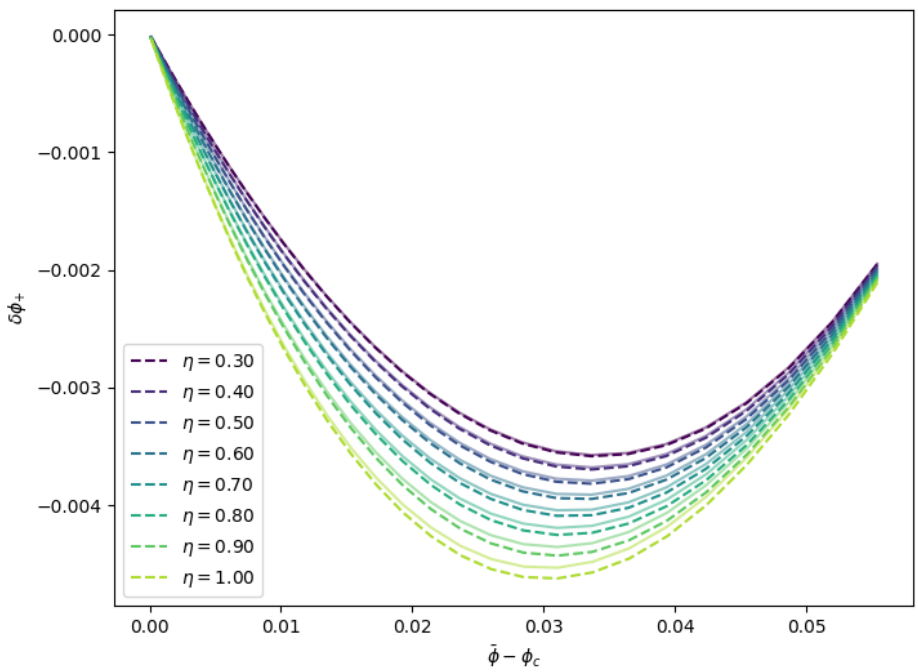

Figure 4: Comparison between numerical solution of EL equations (solid lines) and analytical approximation (dashed lines) for $\chi_{12} = 2.3$.

3.3.3 Surface tension approximation with interfacial absorption

Next we can look at the interface width, $w = \Delta\phi/\nabla\phi_{-,IF}$, where $\nabla\phi_{-,IF}$ is the gradient of ϕ_- at the interface center. At the interface we have $\phi_- = \nabla\phi_+ = 0$. We can use the conserved quantity from Eq. 17 to rewrite this as

$$f - 2\mu\phi_+ + \Pi = 2\phi_+ \ln(\phi_+) + (1-2\phi_+)\ln(1-2\phi_+) - \frac{\phi_+^2}{\phi_s} - 2\mu\phi_+ + \Pi = \frac{(\nabla\phi_-)^2}{\bar{\phi}_c}$$

$$\begin{aligned}\phi_-'(0) = \Bigg\{ \bar{\phi}_c \Big[& 2\phi_{+,0}\ln(\phi_{+,0}) + (1-2\phi_{+,0})\ln(1-2\phi_{+,0}) - \frac{\phi_{+,0}^2}{\phi_s} \\ & - (\bar{\phi}+\Delta\phi)\ln(\bar{\phi}+\Delta\phi) - (\bar{\phi}-\Delta\phi)\ln(\bar{\phi}-\Delta\phi) \\ & - (1-2\bar{\phi})\ln(1-2\bar{\phi}) + \frac{\bar{\phi}^2}{\phi_s} + \frac{\Delta\phi^2}{\bar{\phi}_c} \\ & - 2(\phi_{+,0}-\bar{\phi})\left(-\frac{\bar{\phi}}{\phi_s} - \frac{\Delta\phi}{\bar{\phi}_c} + \ln(\bar{\phi}+\Delta\phi) - \ln(1-2\bar{\phi})\right)\Big]\Bigg\}^{1/2}\end{aligned} \tag{30}$$

and thus:

$$w = \sqrt{\frac{\bar{\phi}_c(\Delta\phi)^2}{2\phi_+\ln(\phi_+) + (1-2\phi_+)\ln(1-2\phi_+) - \frac{\phi_+^2}{\phi_s} - 2\mu\phi_+ + \Pi}} \tag{31}$$

The analytical approximation for the interface width can be seen in figure 5. Here we can see that including the absorption yields a much better agreement with the numerical results.

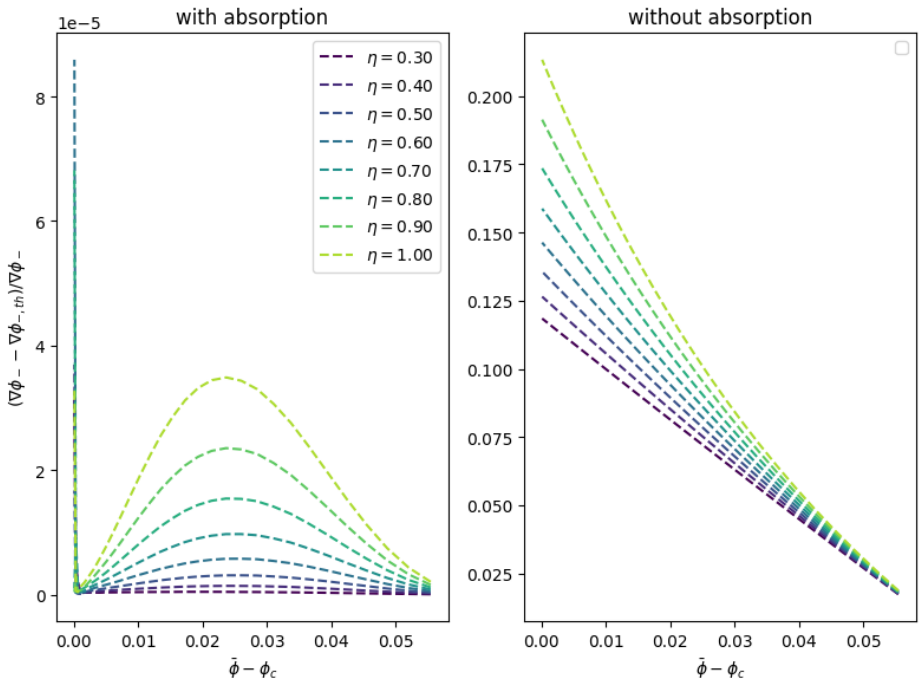

Figure 5: Relative difference between numerically and analytical approximation of the gradient of $\phi_-(0)$ with (left) and without (right) absorption for $\chi_{12} = 2.3$ and varying η.

4 Fast Binodal Construction

In the previous chapter, we investigated the interfacial profiles of ternary systems. For faster convergence of the equilibrium profiles, we initialized the system using knowledge of the bulk compositions of the coexisting phases. These compositions can be obtained in a timely manner using the flory python package. This was initially the method of choice. However, after optimizing the numerical schemes for the interfacial profiles precomputing the bulk compositions of the coexisting phases became the bottleneck of the entire procedure. Since we are interested in systems for ever increasing mean volume fraction, we are essentially asking for the binodal curve Since the system is equilibrated iteratively for ever increasing mean volume fraction of the second species, one can ask the question of using information of the prior iteration to converge faster to the next bulk compositions. In this section we will present a method which allows for a rapid computation of the binodal in phase space for the ternary system.

4.1 Tracing out the binodal curve in phase space

Add a schematic figure showing the binodal curve with points on it representing previous iterations and the step along the curve (tangential step) and a projection step to the binodal. Should this schematic be for dilute and dense phase or just one of them? In order to better understand this algorithm it is helpful to first think about the number of degrees of freedom and the number of constraints that define the binodal curve in phase space. For two distinct phases in the ternary incompressible system, we have four degrees of freedom, which are the two independent volume fractions in each of the two phases. On the other hand, the coexistence conditions impose three constraints. This means that we have one degree of freedom left, which implies that the binodal curve is a one-dimensional manifold in the four-dimensional space of phase compositions.

4.1.1 Coexistence conditions

Given two phases in a ternary system a point on the binodal is uniquely given by the coordinates in phase space:

$$\boldsymbol{\phi} = (\phi_1^{(1)}, \phi_2^{(1)}, \phi_1^{(2)}, \phi_2^{(2)})^T$$

, where the superscript denotes the phase and the subscript denotes the species. These points are not arbitrary but are constrained by the coexistence conditions, which require that the chemical potentials and osmotic pressure of the two phases are equal.

$$\mu_i(\phi_1^{(1)}, \phi_2^{(1)}) = \mu_i(\phi_1^{(2)}, \phi_2^{(2)})$$
$$\Pi(\phi_1^{(1)}, \phi_2^{(1)}) = \Pi(\phi_1^{(2)}, \phi_2^{(2)})$$

For the Flory-Huggins free energy, the chemical potentials and osmotic pressure can be expressed as

$$\mu_i(\phi_1, \phi_2) = \frac{\partial f}{\partial \phi_i} = \ln(\phi_i) - \ln(\phi_0) + \sum_{j=1}^{2} \chi_{ij}\phi_j$$

$$\Pi(\phi_1, \phi_2) = -f + \sum_i \mu_i \phi_i = -\ln(\phi_0) + \tfrac{1}{2} \sum_{i,j=1}^{2} \phi_i \chi_{ij} \phi_j$$

The algorithm we will present in the next sections relies on the fact that we start with a point on the binodal, i.e. a point that satisfies the coexistence conditions.

4.1.2 Tangential step along the binodal

Given that we know the bulk compositions of the coexisting phases for a given point in phase space, can we move a small step in phase space and remain on the binodal curve? For a step that is small enough, we can expand the coexistance conditions to first order in the step size $\delta\boldsymbol{\phi} = (\delta\phi_1^{(1)}, \delta\phi_2^{(1)}, \delta\phi_1^{(2)}, \delta\phi_2^{(2)})^T$:

$$\boldsymbol{E}(\boldsymbol{\phi} + \delta\boldsymbol{\phi}) = \boldsymbol{E}(\boldsymbol{\phi}) + \boldsymbol{J}\,\delta\boldsymbol{\phi} + \mathcal{O}(\delta\boldsymbol{\phi}^2) = \boldsymbol{0} \tag{32}$$

Where $\boldsymbol{E} = (\mu_1^{(1)} - \mu_1^{(2)}, \mu_2^{(1)} - \mu_2^{(2)}, \Pi^{(1)} - \Pi^{(2)})^T$ and $\boldsymbol{J}$ is the Jacobian matrix of the system:

$$\boldsymbol{J} = \begin{pmatrix} \frac{\partial \mu_1^{(1)}}{\partial \phi_1^{(1)}} & \frac{\partial \mu_1^{(1)}}{\partial \phi_2^{(1)}} & -\frac{\partial \mu_1^{(2)}}{\partial \phi_1^{(2)}} & -\frac{\partial \mu_1^{(2)}}{\partial \phi_2^{(2)}} \\ \frac{\partial \mu_2^{(1)}}{\partial \phi_1^{(1)}} & \frac{\partial \mu_2^{(1)}}{\partial \phi_2^{(1)}} & -\frac{\partial \mu_2^{(2)}}{\partial \phi_1^{(2)}} & -\frac{\partial \mu_2^{(2)}}{\partial \phi_2^{(2)}} \\ \frac{\partial \Pi^{(1)}}{\partial \phi_1^{(1)}} & \frac{\partial \Pi^{(1)}}{\partial \phi_2^{(1)}} & -\frac{\partial \Pi^{(2)}}{\partial \phi_1^{(2)}} & -\frac{\partial \Pi^{(2)}}{\partial \phi_2^{(2)}} \end{pmatrix}$$

The first term on the right-hand side of Eq. 32 vanishes since $\boldsymbol{\phi}$ is a point on the binodal. In order to remain on the binodal, we must therefore have

$\boldsymbol{J}\,\delta\boldsymbol{\phi} = \boldsymbol{0}$. This means that the step along the binodal curve is proportional to the null space of the Jacobian matrix $\delta\boldsymbol{\phi} \propto \text{null}(\boldsymbol{J})$.

4.1.3 Projecting back onto the coexistence manifold

Even though this is a good start it still introduces an error of order $\delta\boldsymbol{\phi}^2$ which is propagated across iterations and can lead to a significant deviation from the binodal curve after many iterations. Although one can reduce this error by taking smaller steps, this also increases the number of iterations required to trace out the binodal curve, which is not ideal. Instead we can ask a slightly different question: given that we are close but not exactly on the binodal, in which direction should we step in order be at the binodal? Mathematically, this can be expressed the same as Eq. 32, but now the first term on the right-hand side does not vanish and we need to solve for $\delta\boldsymbol{\phi}$ such that the entire right-hand side vanishes. This can be written as

$$\boldsymbol{J}\,\delta\boldsymbol{\phi} + \boldsymbol{E}(\boldsymbol{\phi}) = \boldsymbol{0}$$

It is easy to see that the soltution to this equation is given by the pseudo-inverse of the Jacobian:

$$\delta\boldsymbol{\phi} = -\boldsymbol{J}^{+}\boldsymbol{E}(\boldsymbol{\phi})$$

where $\boldsymbol{J}^{+} = \boldsymbol{J}^{T}(\boldsymbol{J}^{T}\boldsymbol{J})^{-1}$. The Eq. 32 is invariant under translations of $\delta\boldsymbol{\phi}$ along the null space of the Jacobian. Since the pseudo-inverse solution is orthogonal to the null space it minimizes the step size, by which we ensure that the tangential step and the projection step are independent and do not interfere with each other. This will become important when during algorithmic implementation we need to define a step size for the tangential step, which is not well defined if the projection step also moves tangentially along the binodal curve.

The pseudo-inverse is easy to compute using singular value decomposition (SVD) of the Jacobian

$$\boldsymbol{J} = \boldsymbol{U}\,\boldsymbol{\Sigma}\,\boldsymbol{V}^{T}$$

where $\boldsymbol{U}$ and $\boldsymbol{V}$ are orthogonal matrices and $\boldsymbol{\Sigma}$ is a diagonal matrix containing the singular values of the Jacobian. The pseudo-inverse can then be computed as

$$\boldsymbol{J}^{+} = \boldsymbol{V}\,\boldsymbol{\Sigma}^{+}\,\boldsymbol{U}^{T}$$

where $\boldsymbol{\Sigma}^{+}$ is obtained by taking the reciprocal of the nonzero singular values. Vanishing singular values of the Jacobian correspond to directions which the

Jacobian projects to zero. This mean that instead of trying to reconstruct this lost information with the pseudo-inverse, one simply sets the reciprocal of the zero singular values to zero. In practice, one needs to also consider small but nonzero singular values, which can lead to numerical instability. Therefore, one needs to set a threshold below which the reciprocal of the singular value is set to zero.

Once one has computed the projection step, one can project back onto the binoda curve. Because second order errors are neglected the projection step is not perfect and might require multiple iterations to converge to the desired accuracy. Most importantly, the projection step is only guaranteed to converge if the initial point is close enough to the binodal curve, which is the case if we take small enough tangential steps. All in all the projection step allows us to take much larger tangential steps without introducing significant errors, which allows us to trace out the binodal curve much faster and yet more accurately than without the projection step.

4.2 Algorithm Considerations

4.2.1 Initialization of the stepping scheme

This algorithm relies on the fact that we have a point on the binodal to start with. In our implementation we consider two fundamentally different types of initial points: the dilute limit of one species, where the system effectively reduces to a binary system, and the critical point, which is a distinct thermodynamic state rather than a dilute-limit construction. Both types of initial points can be found analytically for the Flory-Huggins free energy, which makes them ideal candidates for initialization.

For explaining reasons we will consider the dilute limit of the zeroth species, but the same logic applies to the other species by relabeling the volume fractions. If the repelling interaction of the two remaining species is strong enough ($\chi_{12} > 2$) then one can find the coexisting phases for the binary system from the coexistence condition:

$$\ln(\phi) - \ln(1-\phi) + \chi_{12}(1-2\phi) = 0$$

Where ϕ is the volume fraction of one of the species, and the other species is given by $1-\phi$. The dilute limit is then given by

$$(\phi_1^{(1)}, \phi_2^{(1)}, \phi_1^{(2)}, \phi_2^{(2)}) = (\phi - \epsilon, 1 - \phi - \epsilon, 1 - \phi - \epsilon, \phi - \epsilon)$$

such that $\phi_0^{(1)} = \phi_0^{(2)} = 2\epsilon$ is small, which allows us to be close enough to the binodal to be able to project back on the binodal and start the stepping scheme. Because the null space of the Jacobian is invariant under sign change one needs to choose the direction of the tangential step away from the border.

The critical point can also be found analytically by finding the point in phase space where the second and third derivatives of the free energy vanish (see Section **??**). Because multiple singular values of the Jacobian nearly vanish near the critical point, it can be challenging to compute the tangential direction of the binodal curve from the Jacobian. However because the spinodal and binodal curves overlap at the critical point one can deduce that the tangential direction of the binodal curve at the critical point is given by the null vector of the Hessian of the free energy.

4.2.2 Handling binodal branching events

While developing this algorithm we found a curious case for the symmetric ternary system **??** where the spinodal crossed the binodal, which would mean that part of the binodal was locally unstable but globally stable, which is illogical. This necessetated the existence of additional binoda branches enclosing the spinodal curve. After investigating the the singular values of the Jacobian throughout the binodal curve, points appeared where the second smalles singular value also vaninshed, which means that the null space of the Jacobian is degenerate and one can step in multiple directions, which corresponds to the branching of the binodal curve. Therefore in order to get the entire binodal curve one needs to track the two smalles singualr values of the Jacobian to detect these branching events and then step in both directions.

4.3 Implementation

The phase diagram of a ternary system in the Flory-Huggins theory is uniquely determined by the interaction parameters χ_{ij}. For our algorithm this has the effect that the user only needs to provide the interaction parameters and the algorithm will compute the entire phase diagram. All other parameters only define the numerical details which can be tuned to optimize the performance of the algorithm but do not affect the results in a significant way as long as they are within reasonable ranges.

4.3.1 Computing the spinodal and binodal curve

Once the interaction parameteres are provided, one first computes the spinodal curve and critical points, which can be done analytically for the Flory-Huggins free energy. Starting from an initial point, one can step along the binodal curve by repeating the steps:

- Take a tangential step along the binodal curve
- Project back onto the binodal curve using the pseudo-inverse of the Jacobian

The stepping is repeated until one of three termination criteria is met, which are:

- The volume fraction of one of the species is outside of the physically meaningful range, which is between 0 and 1.
- We reach a critical point, which is implemented by terminating whenever the distance between the two phases is smaller than a certain threshold.
- In rare cases the binodal can loop back on itself, which can lead to infinite loops. To prevent this we monitor the distance between the current point and the initial point and terminate if this distance is smaller than a certain threshold.

During the stepping scheme, one needs to monitor the singular values of the Jacobian to detect branching events. Whenever the second smallest singular value is smaller than a certain threshold A new initial point is added to a task list. This initial point includes location in phase space and the direction is not being explored by the current process. After all initial points from the dilute limit and critical points are explored, the algorithm continues by picking new initial points from the task list until the task list is empty. In order to avoid compute sections with the same initial point, we cache the initial points that have already been explored and skip them when they are picked from the task list.

4.3.2 Post processing

After all binodal branches have been computed, one needs to remove duplicates, which can occur if one for example starts at the boundary of the phase diagram and the critical point of the same binodal section. Once all duplicates are removed, one can look for intersecting points of two branches of the binodal, as this is where two tie lines go in different directions, which means that we have three coexisting phases. Once the algorithm has completed we are left with a set of spinodal and binodal branches, as well as the dividing lines between the one phase, two phase and three phase regions in phase space.

4.3.3 Numerical details

Due to the discrete nature of the stepping scheme we often times rely on numerical threshold values throughout the algorithm. For the step size of the tangential step, we typically choose a static value of around 0.01. In the case where the second smalles singular value of the Jacobian is smaller than 10^{-1} we adapt the step size to be proportional to the singular value which is capped from below by a value of around 10^{-4} to prevent the step size from becoming too small. This is done to better resolve the branching events. The branching events are triggered when the second smallest singular value of the Jacobian is smaller than around 10^{-3}. Additionally, for the projection loop we typically choose a threshold of around 10^{-10} for the norm of $\boldsymbol{E}(\boldsymbol{\phi})$ to determine whether we are close enough to the binodal curve. The termination criteria for coexistence points near the critical point is

$$|\boldsymbol{\phi}^{(1)} - \boldsymbol{\phi}^{(2)}| < 10^{-3}$$

In order to facilitate numerical stability of the projection step near the critical point we set all singular values of the Jacobian that are smaller than 10^{-6} to zero.

Numerical Values

- 0.01: typical static step size for the tangential step.
- 10^{-1}: threshold below which the second smallest singular value triggers adaptive step-size scaling.
- 10^{-4}: lower cap for the adaptive step size.
- 10^{-3}: threshold for triggering branching events (based on the second smallest singular value).
- 10^{-10}: projection-loop threshold for $\|\boldsymbol{E}(\boldsymbol{\phi})\|$ to determine proximity to the binodal curve.
- 10^{-3}: termination criterion near the critical point, $|\boldsymbol{\phi}^{(1)} - \boldsymbol{\phi}^{(2)}| < 10^{-3}$.
- 10^{-6}: cutoff below which Jacobian singular values are set to zero near the critical point.

4.4 Results and validation

4.4.1 Validation of the algorithm

- Validate by equilibrating spacially extended systems for randomly sampled points in phase space and comparing the bulk compositions of the coexisting phases with the results from the binodal construction.
- Validate by comparing with flory package.

4.4.2 Identification of three-phase regions

- According to Gibbs phase rule, in a ternary system we can have up to three coexisting phases.
- One can detect three phase regions by looking at the intersecting points of two branches of the binodal because at that point we have two tie lines going in different directions.
- It can also happen that the a three phase region exists inside of another three phase region. In this case

4.4.3 Case study: symmetric ternary system

- symmetric ternary systems ($\chi_{01} = \chi_{02} = \chi_{12} = \chi$) offer a rich number of interesting features as one increases χ such as branching bifurcations. Up to nine critical points. Binodal branches which loop back on themselves. Three three phase regions which join to a single three phase region, tie lines which rotate as one moves across phase space, etc.

- Verify with results from the literature

4.4.4 Other interesting cases

- Binodal not crossing boundary, if flory parameter are negative enough (strong attraction)

- Systems with binodals but without critical points